job_6-Evaluation report

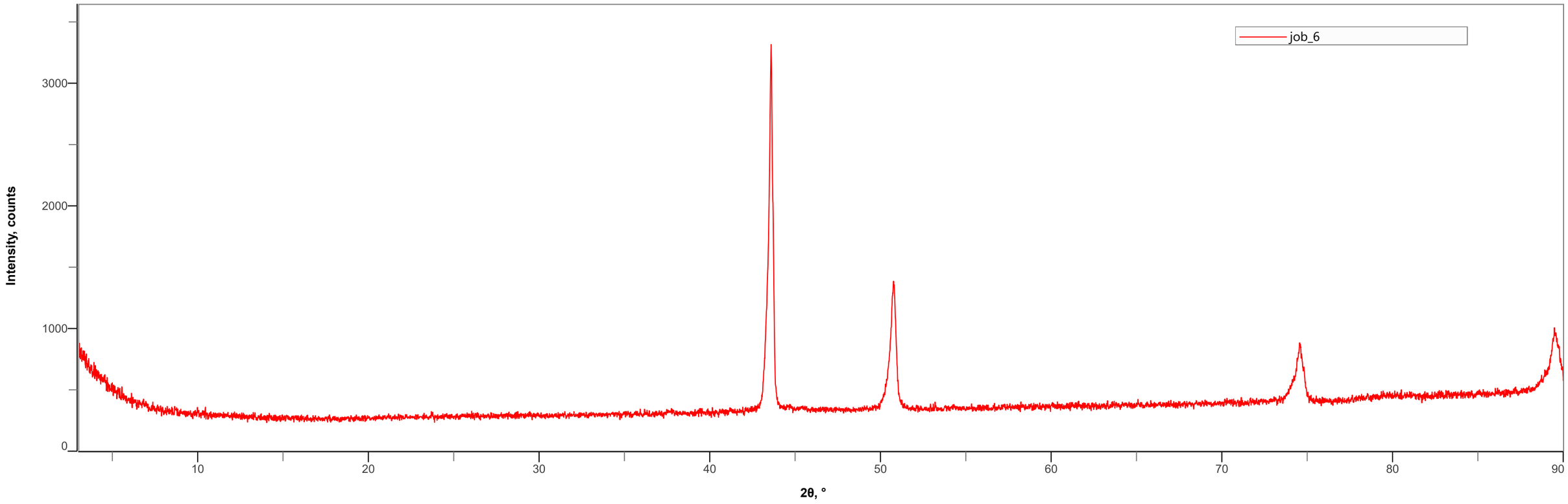
General information

Analysis date	2025-11-27 13:29:46	Measurement start time	2025-11-23 10:15:46
Analyst	Administrator	Operator	Administrator
Sample name	job_6	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_6.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

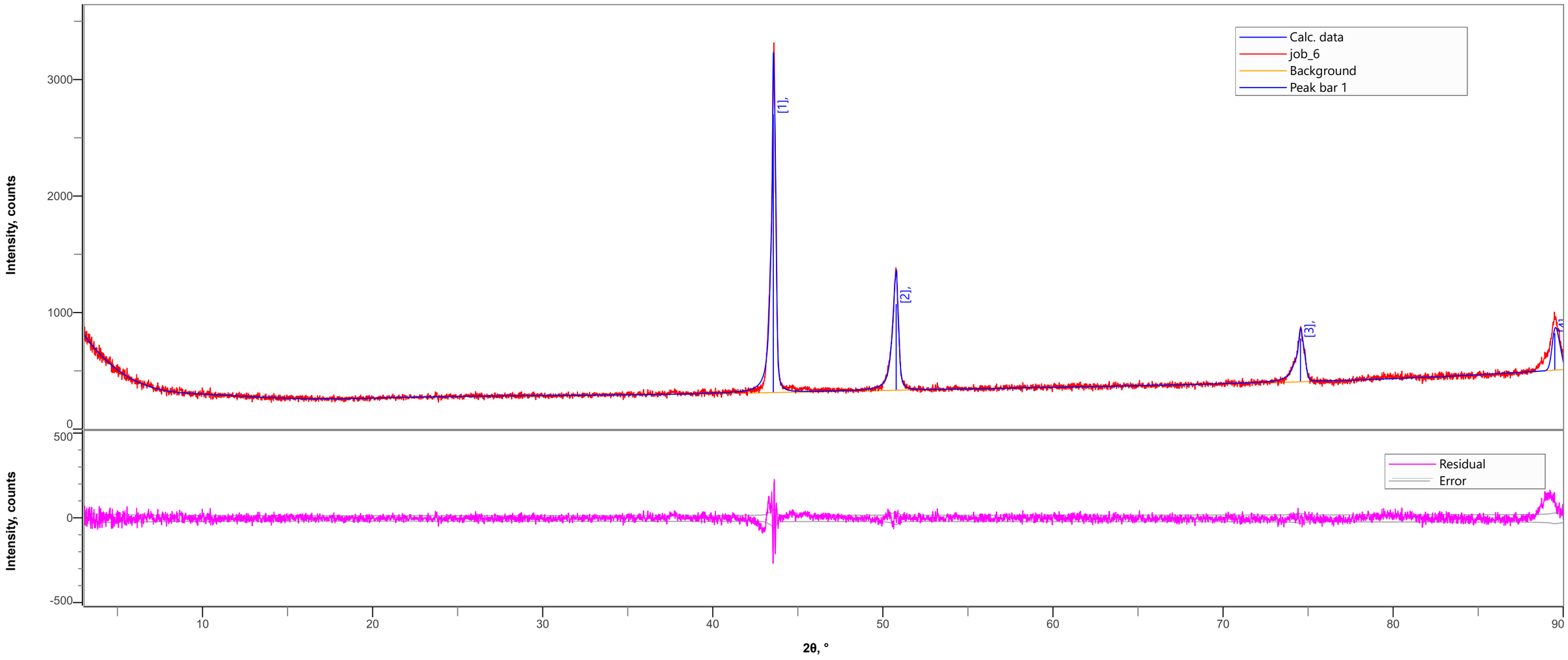
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

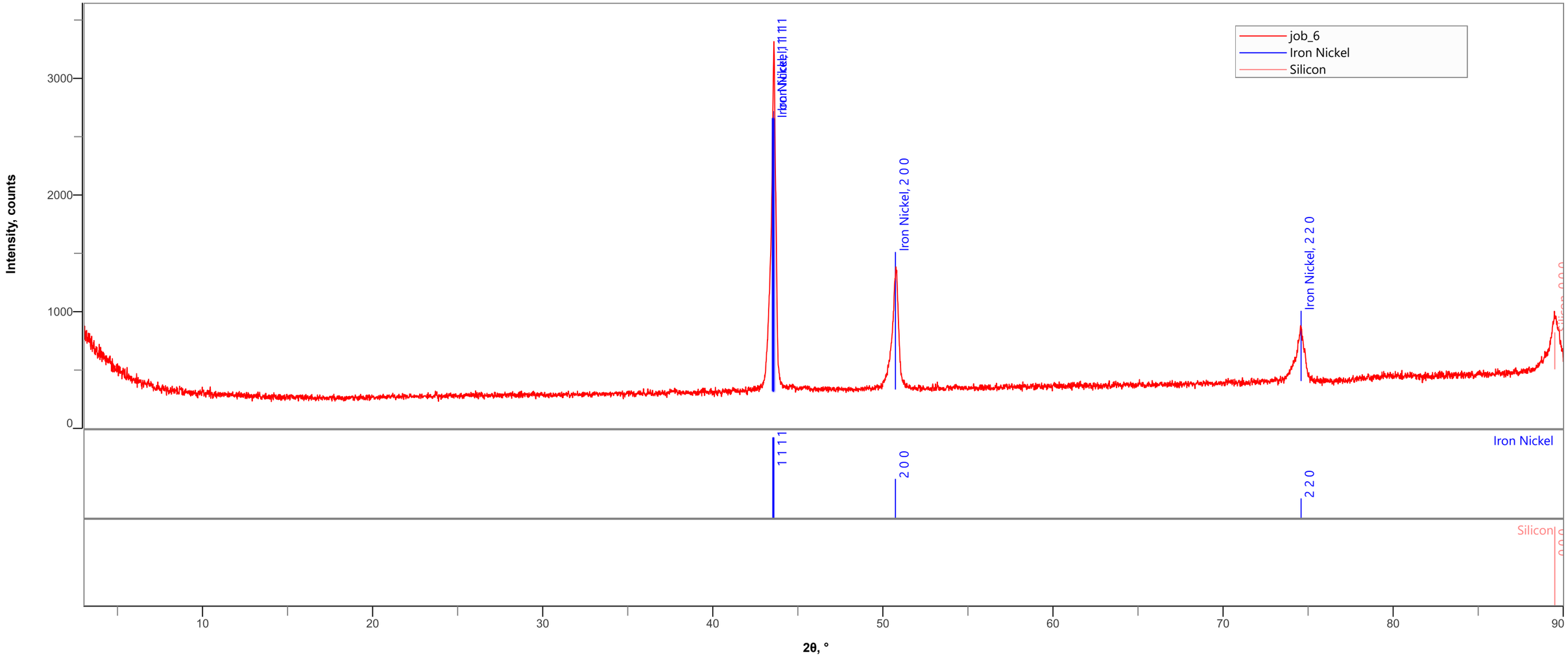
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Iron Nickel	Fe0.64 Ni0.36	0.187	S/M:PDF-5+ 2026	225 : Fm-3m	00-047-1405
Silicon	Si	0.501	S/M:PDF-5+ 2026		00-047-1187

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	43.5358(12)	2.07712(5)	2384(44)	0.184(2)	732(3)	0.307(7)	0.90(2)	1.550(19)	0.47(2)	485(5)	Iron Nickel: 1 1 1	Fe0.64 Ni0.36	00-047-1405	100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	50.754(5)	1.79734(17)	737(20)	0.347(5)	359(2)	0.487(17)	2.8(2)	0.70(3)	0.87(8)	265(4)	Iron Nickel: 2 0 0	Fe0.64 Ni0.36	00-047-1405	49.01	Split Pseudo-Voigt	-	-	-	-	-	
3	74.508(6)	1.27248(9)	372(13)	0.334(11)	212(3)	0.57(3)	1.26(11)	1.53(5)	0.38(9)	312(10)	Iron Nickel: 2 2 0	Fe0.64 Ni0.36	00-047-1405	28.90	Split Pseudo-Voigt	-	-	-	-	-	
4	89.454(11)	1.09460(11)	316(10)	0.40(3)	134(18)	0.42(7)	1.03(9)	0.00(16)	0.0(3)	295(19)	Silicon	Si	00-047-1187	18.30	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Iron Nickel	3.59755	3.59755	3.59755	90.000	90.000	90.000
Silicon	0.00000	0.00000	0.00000	90.000	90.000	90.000

d-I List

Iron Nickel

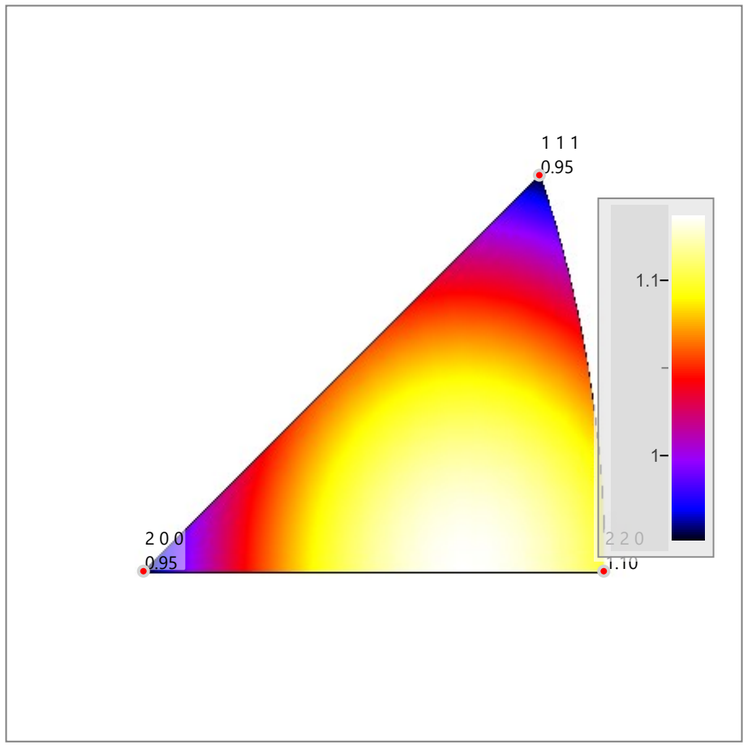
No.	2θ, °	d, Å	h k l	Norm. I.
1	43.53746	2.07705	1 1 1	100.00
2	50.71101	1.79878	2 0 0	45.10
3	74.54602	1.27193	2 2 0	22.55

Silicon

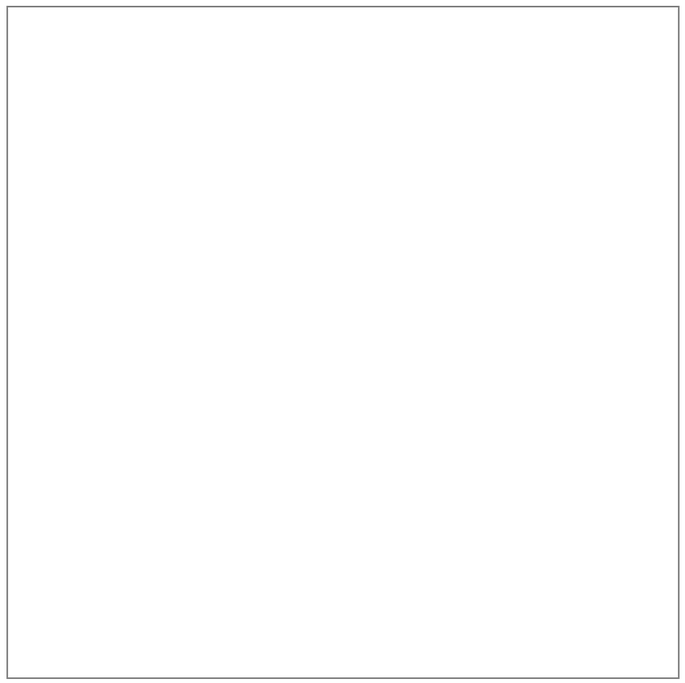
No.	2θ, °	d, Å	h k l	Norm. I.
1	89.61940	1.09300	0 0 0	100.00

Simple Inverse Pole Figure

Iron Nickel

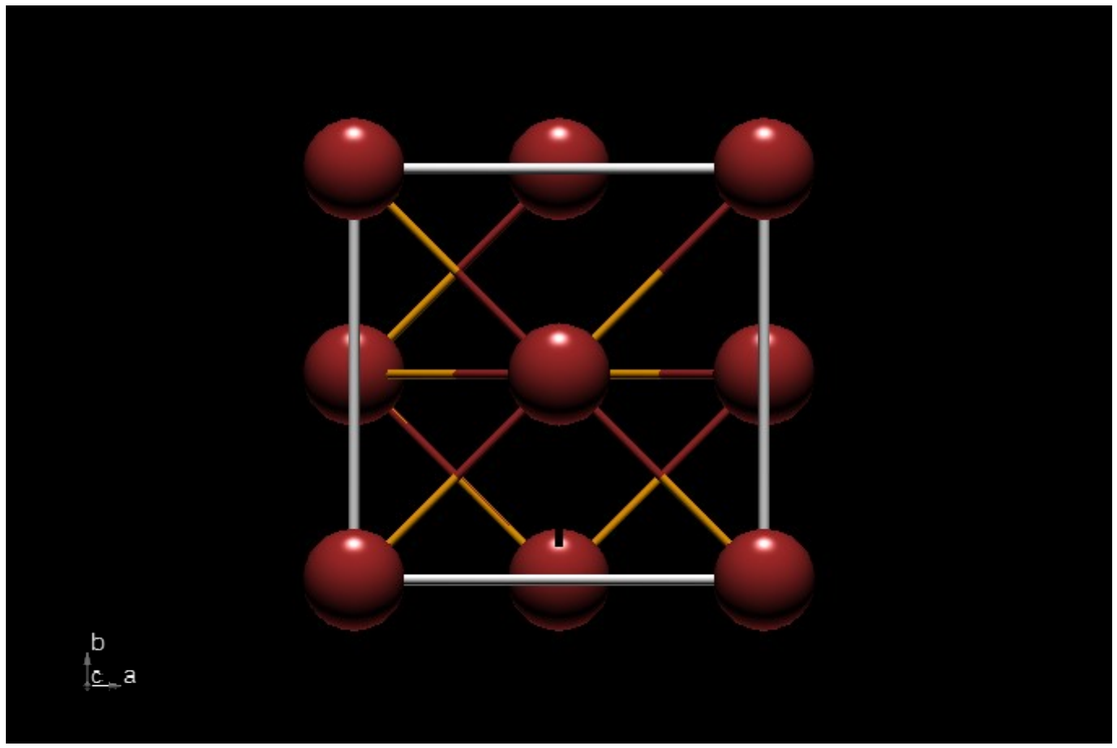


Silicon



Crystal Structure

Iron Nickel



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Fe1(Fe)	0.000000	0.000000	0.000000	0.640	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.360	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Fe1-Ni1(x,1/2+y,1/2+z)	2.54385
Fe1-Ni1(x,-1/2+y,1/2+z)	2.54385
Fe1-Ni1(x,1/2+y,-1/2+z)	2.54385
Fe1-Ni1(1/2+x,1/2+y,z)	2.54385
Fe1-Ni1(-1/2+x,1/2+y,z)	2.54385
Fe1-Ni1(1/2+x,-1/2+y,z)	2.54385
Fe1-Ni1(1/2+x,y,1/2+z)	2.54385
Fe1-Ni1(-1/2+x,y,1/2+z)	2.54385
Fe1-Ni1(1/2+x,y,-1/2+z)	2.54385
Fe1-Fe1(x,1/2+y,1/2+z)	2.54385
Fe1-Fe1(x,-1/2+y,1/2+z)	2.54385

Distance

Bond Def.	Distance, Å
Fe1-Fe1(x,1/2+y,-1/2+z)	2.54385
Fe1-Fe1(1/2+x,1/2+y,z)	2.54385
Fe1-Fe1(-1/2+x,1/2+y,z)	2.54385
Fe1-Fe1(1/2+x,-1/2+y,z)	2.54385
Fe1-Fe1(1/2+x,y,1/2+z)	2.54385
Fe1-Fe1(-1/2+x,y,1/2+z)	2.54385
Fe1-Fe1(1/2+x,y,-1/2+z)	2.54385